
Association rule mining and Apriori Exercises

Exercise 1

Consider the itemsets that can be generated using $I = \{A, B, C, D, E\}$:

- How many possible frequent itemsets (excluding the empty set) can we have? How would this number change if we had 6 items in I , i.e., $\{A, B, C, D, E, F\}$?

$$2^6 - 1 = 63$$

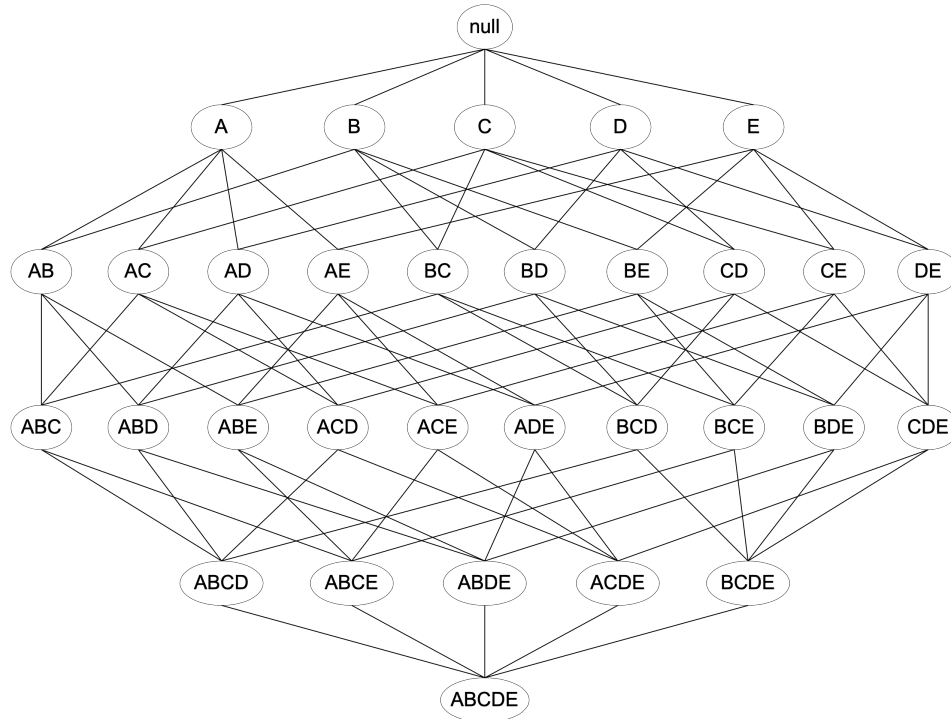
- If the itemset BE is found to be infrequent, then which are the sets that will be pruned based on the Apriori principle and why?

Apriori says $s(x) \geq s(y)$ if x is subset of y . So if x is infrequent then all its supersets are pruned. ex. ~~$= \{ABDE, BCE, \text{etc}\}$ all sets including BE basically~~

Exercise 2

Consider now the following transaction database D and the lattice of itemsets that can be generated using $I = \{A, B, C, D, E\}$:

Transaction ID	Items
1	$\{A, B\}$
2	$\{B, C, D\}$
3	$\{A, B, C, D\}$
4	$\{A, B, D\}$
5	$\{A, B, C, E\}$



- What is the frequency of each possible itemset? Please annotate the frequency clearly on the lattice above.

Frequency $\{A\} = 4/5$ (according to transactions) $\{ABC\} = 2/5$

- If the *absolute support* threshold is equal to 2, which itemsets are maximal frequent and why? Please indicate them in the lattice above.

Support of $\{ABC\} = 2/5$ maximal (its superset $abcd$, $abce$ are not frequent) : $\{BCD\} = \{2/5\}$ (its superset are not frequent support < 2)

- Which itemsets are closed and why? Please indicate them in the lattice above.

Closed itemset are ones if none of its immediate superset does not have same support as them.

$\{ABC\}, \{BCD\} \rightarrow$ Maximal itemsets are closed (since all their supersets are not frequent therefore not have same support as them), $\{AB\}, \{BD\}$

- What is the practical usefulness of closed itemsets compared to maximal itemsets? Please explain in one sentence.

It helps in reconstructing set of frequent items along with their frequencies.

Exercise 3

Consider the following transaction database D. Each transaction contains a set of items:

Transaction ID	Items
1	{A, B, C}
2	{B, D}
3	{A, B}
4	{B, C, D}
5	{B, C}
6	{A, B, C}
7	{A, C, D}
8	{A, C}

- Calculate and report the absolute support for every possible itemset that can be generated using the universe $I = \{A, B, C, D\}$, based on the transactions in the database. Identify and report the frequent itemsets based on a minimum absolute support threshold of 2.

always go step
by step 1st 1
element
itemset

all single items
have support $\geq$
2 $\rightarrow$ all are
frequent.

{A,B,C}: T1,
T6 $\rightarrow$ count =
2 

AB, BD, BC, AC, BC, ≥ 2 SUPPORT

- Among the frequent itemsets, identify and report the closed itemsets. Explain in one sentence why you chose them.

{A}, {B}, {C},
{D}, {A,B},
{A,C}, {B,C},
{B,D}, {C,D},
{A,B,C}

1st single items A,B,C, $\rightarrow$ then 2 itemset $\rightarrow$ aise jao aage

AB/AC = 3 ABC = 2 SUPPORT SO CLOSE ITEMSETS IF NONE OF ITS SUPERSET HAS SAME SUPPORT

BD = 2 below superset < 2 support

BC = 3 below ABC = 2 rest < 2 support

- Among the frequent itemsets, identify and report the maximal itemsets. Explain in one sentence why you chose them.

ABC = 2 all its superset are not frequent

a b c d = all not either they are not frequent (d) or there superset are frequene(a b c)

BD = yes (all below abd bcd are not frequent)

- Explain why certain itemsets are closed but not maximal, and vice versa.

All maximal itemset are closed as maximal itemset are frequent but all its superset are not frequent making it a closed itemset as the support of maximal itemset $>$ support of its superset.

Eg. ABC and BD

While opposite need not to be true, since closed itemset support should not be equal to its superset support, but it can happen that superset support is enough to make it frequent and then its superset's superset may not be frequent like in case of maximal

~~eg. {A,B} is closed because none of its supersets have support = 3, but {A,B,C} is frequent, so {A,B} is not maximal.~~

Exercise 4

(a) Describe a way of using the Apriori algorithm for mining infrequent (meaning, very rare) itemsets. Note that you do not need to change the algorithm itself, but simply come up with a way of using it with the appropriate minsup tuning.

X IS NOT FREQUENT THEN ALL ITS SUPERSET ARE INFREQUENT USING THIS IF WE FIND SUCH X THEN A NUMBER OF POSSIBLE INFREQUENT ITEMSETS CAN BE FOUND.

(b) Explain briefly how does the Apriori algorithm prune infrequent itemsets, and why is this pruning step essential for efficient frequent itemset mining.

$S(X) \geq S(Y)$ IF X SUBSET OF Y SO IT ALSO MEANS THAT IF X IS NOT FREQUENT ITEMSET THEN ALL THE SUPERSET FROM X ARE PRUNED AS THEY CAN NEVER BE FREQUENT ITEMSET

Apriori generates candidate itemsets level by level (1-itemsets, 2-itemsets, etc.).

Before counting support for larger itemsets, it prunes candidates whose subsets are infrequent.

Exercise 5

Suppose that you have used the Apriori algorithm (using a relative support threshold of minsup) and have found all frequent itemsets from a given transactional dataset from the supermarket. Amongst the frequent itemsets you have discovered, you have the following itemsets and relative supports (indicated next to each itemset):

- {cheese}: 50%
- {ham}: 50%
- {bread}: 100%
- {bananas}: 50%
- {bread, bananas}: 25%

If the database contains 100 transactions, answer the following questions based on the above information:

1. How many customers bought cheese?

50

50

2. Given the additional information that all customers who buy “ham” also buy “cheese”, what is the relative support of {ham, cheese}?

$\text{supp}(\{\text{ham}, \text{cheese}\}) = \text{supp}(\{\text{ham}\}) = 50\%$

database mein
jaha ham hai
waha socho
ham,cheese
aa gaya hai
now

Since every transaction containing ham also contains cheese,

supp

(

4

{

h

a

m

,

c

-
3. If we remove all transactions where “bread” occurs, what fraction of the database will be left?

0

4. What is the confidence of bread $\rightarrow$ bananas?

Recall the definition of confidence: confidence ($X \rightarrow Y$) = $\text{supp}(X, Y) / \text{supp}(X)$
25 / 100